The triangle inequality

1

a

i

- $7 + 12 > 17$
- $7 + 17 > 12$
- $17 + 12 > 7$

ii Yes

b

i

- $3.2 + 7.4 > 9.6$
- $3.2 + 9.6 > 7.4$
- $9.6 + 7.4 > 3.2$

ii Yes

c

i

- $9 + 4 > 5$
- $9 + 5 > 4$
- $5 + 4 = 9$

ii Yes

d

i

- $3.1 + 7.7 < 11.8$
- $3.1 + 11.8 > 7.7$
- $11.8 + 7.7 > 3.1$

ii No

2

a $8 + 8 > 8$

b Yes

3

a Yes **b** No **c** No

4 The angle sum exceeds 180 degrees.

5

a $9 < a < 43$ **b** $7.4 < a < 41$ **c** $7 < a < 35$ **d** $17.6 < a < 75.8$

e $12 < a < 44$ **f** $0.8 < a < 4$ **g** $10 < a < 44$ **h** $7 < a < 43$

Additional questions

6 The sum of the lengths of any two sides of a triangle is greater than the length of the third side.

$AB + BC > AC$ and $AB + AC > BC$ and $AC + BC > AB$

7

a Valid

b Not valid, $4 + 7 \not> 12$

c Not valid, $6 + 7 \not> 13$

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8

a

i $17 + 26 > a$, so $43 > a$
$17 + a > 26$, so $a > 9$
$26 + a > 17$, so $a > -9$, which is true if $a > 9$

ii $9 < a < 43$

b

i $15 + 18 > a$, so $33 > a$
$15 + a > 18$, so $a > 3$
$18 + a > 15$, so $a > -3$, which is true if $a > 3$

ii $3 > a > 33$

9 $-\frac{3}{5} < x < 11$

10 He is correct that a valid triangle does not exist. The two doors just touching and forming a triangle would be a side length of $2 \cdot 36 = 72$ inches. The other two sides are 16 inches and $4.5 \cdot 12 = 54$ inches. However, $16 + 54 \not> 72$, so this is not a valid triangle.
His error is that just because they do not form a triangle, we cannot conclude that the doors do not touch. We can give a few different reasons for this:

- Although, the range of values that forms a triangle is $38 < x < 70$, this is a right triangle, so there is actually only one valid triangle with hypotenuse length of 56.32- which can be calculated with the Pythagorean theorem
- The doors will not ever touch if the sum of the two doors is less than the hypotenuse, which is not the case
- Since it is over the hypotenuse, the two doors will touch and form a quadrilateral

11 This is not a valid conclusion. This incorrectly assumes that Jacksonville, Orlando and Miami are all collinear. This is not necessarily the case, so they would form a triangle with vertices J for Jacksonville, O for Orlando, and M for Miami. Using the triangle inequality so we know that:
$JO + OM > JM$ or that $OM > JM - JO$, so the distance from Orlando to Miami is greater than 204 miles.

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